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BATPING: Acoustic characterization of thermal gradients along the optical path

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PRISE
PROGRAM FOR RESEARCH IN SCIENCE AND ENGINEERING

Introduction

Image Quality and Precision Cosmology at LSST

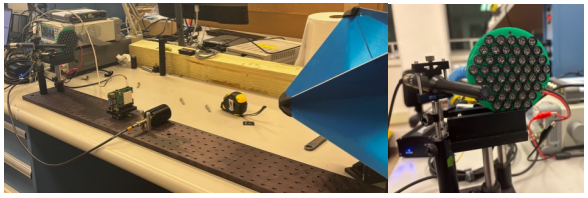
The Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory generates 20TB of astronomical imagery nightly, enabling precision cosmological measurements of dark energy and dark matter. Achieving the survey's science goals requires tight control of image quality throughout the 10-year campaign.

Dome Seeing as Limiting Factor

- Assessments from the first preview of LSST data indicate that telescope point spread function (PSF) exceeds the free-air seeing floor by a median of 0.13" (Rosignoli et al. 2026)
- Dome and mirror seeing, budgeted at 0.08", remain uncharacterized (Urbach et al. 2026)
- Correcting the thermal structure to 0.1°C would bring the PSF within 10% of the diffraction limit.
- Previous work in the Stubbs group address seeing characterization with optical approaches

Enter BATPING

Bidirectional Acoustic Traveltime Pulser for Index Gradients (BATPING) measures path-integrated air temperature via the round-trip travel time of ultrasonic pulses, without disrupting normal telescope operation.



AIM: Achieve **sub-0.1°C per-ping temperature precision** along the Rubin Observatory dome optical path, enabling **real-time characterization of dome seeing** during astronomical observations.

Methods

Results

Share what you discovered or created so far

Usually this is where you would share a larger graphic

Make sure to label all graphs

Conclusion

Describe importance of your findings, in field and broadly.



Next Steps

What's next for the project?

What might be next for you?

Acknowledgements
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References

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